

Natural language statement

Domain

GroundTruth

Theorem 3.4 (disintegration) Let ρ be a sigma-finite measure on $S \times T$, where T is Borel. Then

- (i) $\rho = \nu \otimes \mu$ for a sigma-finite measure $\nu \sim \rho(\cdot \times T) = \rho(S, \cdot)$ and a sigma-finite kernel $\mu : S \rightarrow T$,
- (ii) the μ s are ν -a.e. unique up to normalizations, and they are a.e. bounded iff ρ is sigma-finite,
- (iii) when ρ is sigma-finite and $\nu = \rho(S, \cdot)$, we may choose the μ s to be probability measures on T .

Any sigma-finite measure $\nu \sim \rho(\cdot \times T)$ is called a supporting measure of ρ , and we refer to μ as the associated disintegration kernel.

A kernel $\mu : S \rightarrow T$ is said to be finite if $\mu T < \infty$ for all s in S , s -finite if it is a countable sum of finite kernels, and sigma-finite if it satisfies $\mu T < \infty$ for some measurable function $f > 0$ on $S \times T$, where $f_s = f(s, \cdot)$.

Probability

```
def IsAEBoundedKernel {S T : Type*} [MeasurableSpace S] [MeasurableSpace T]
  (nu : Measure S) (kappa : Kernel S T) : Prop :=
  forall^m s dnu, kappa s univ < inf

def IsSigmaFiniteKernel {S T : Type*} [MeasurableSpace S] [MeasurableSpace T]
  (kappa : Kernel S T) : Prop :=
  exists f : S x T -> R=>0inf, Measurable f /\ (forall s t, 0 < f (s, t))
  /\
  forall s, integral^~ t, f (s, t) dkappa s < inf

theorem kallenberg_3_4_disintegration
  {S T : Type*} [MeasurableSpace S] [MeasurableSpace T]
  [StandardBorelSpace T]
  (rho : Measure (S x T)) [SigmaFinite rho] :
  (exists (nu : Measure S) (kappa : Kernel S T), SigmaFinite nu /\
  IsSigmaFiniteKernel kappa /\
  nu << rho.fst /\ rho.fst << nu /\ nu
  tensor m kappa = rho) /\
  (forall (nu nu' : Measure S) (kappa kappa' : Kernel S T),
  SigmaFinite nu -> IsSigmaFiniteKernel kappa -> IsSigmaFiniteKernel kappa' ->
  nu << rho.fst -> rho.fst << nu -> nu tensor m kappa = rho ->
  SigmaFinite nu' -> IsSigmaFiniteKernel kappa' -> IsSigmaFiniteKernel
  kappa' ->
  nu' << rho.fst -> rho.fst << nu' -> nu' tensor m kappa' = rho ->
  exists c : S -> R=>0inf, Measurable c /\ nu' = nu.withDensity c /\
  forall^m s dnu, kappa s = c s * kappa' s) /\
  ((exists (nu : Measure S) (kappa : Kernel S T), SigmaFinite nu /\
  IsSigmaFiniteKernel kappa /\
  IsSigmaFiniteKernel kappa /\ nu << rho.fst /\ rho.fst << nu /\ nu
  tensor m kappa = rho /\
  IsAEBoundedKernel nu kappa) <=> SigmaFinite rho.fst) /\
  (SigmaFinite rho.fst -> exists kappa : Kernel S T, IsMarkovKernel kappa
  /\ rho.fst tensor m kappa = rho) := by
sorry
```

Theorem 4.23 (moments and Holder continuity, Kolmogorov, Loeve, Chentsov) Let X be a process on $\mathbb{R}^d$ with values in a complete metric space (S, ρ) , such that $E \rho(X_s, X_t)^a \leq C |s - t|^{a(d+b)}$, s, t in $\mathbb{R}^d$, for some constants $a, b > 0$. Then a version of X is locally Holder continuous of order p , for every p in $(0, b/a)$.

For any mapping f between two metric spaces (S, ρ) and (S', ρ') , we define the modulus of continuity $w_f = w(f, \cdot)$ by $w_f(r) = \sup\{\rho(f(s), f(t)) : s, t \text{ in } S, \rho(s, t) \leq r\}$, $r > 0$, so that f is uniformly continuous iff $w_f(r) \rightarrow 0$ as $r \rightarrow 0$. Say that f is Holder continuous of order p , if $w_f(r) \leq C r^p$ as $r \rightarrow 0$. The stated property is said to hold locally if it is valid on every bounded set.

For functions $f, g > 0$, we mean by $f \leq C g$ that $f \leq cg$ for some constant $c < \infty$.

Stochastic processes

```
def IsLocallyHolder {D S : Type*} [PseudoMetricSpace D] [PseudoMetricSpace S]
  [Zero D] (p : R=>0) (x : D -> S) : Prop :=
  forall R : R, 0 < R -> exists C : R=>0, HolderOnWith C p x
  (Metric.closedBall 0 R)

theorem kallenberg_4_23_moments_and_holder_continuity
  {Omega S : Type*} [MeasurableSpace Omega] [MetricSpace S]
  [MeasurableSpace S] [BorelSpace S] [CompleteSpace S]
  {d : N} (mu : Measure Omega) [IsProbabilityMeasure mu]
  (X : (Fin d -> R) -> Omega -> S) (hx : forall t, AEMeasurable (X t) mu)
  (Xa b : R) (ha : 0 < a) (hb : 0 < b)
  (hmoment : exists C : R, 0 <= C /\ forall s t,
  integral^~ omega, ENNReal.ofReal ((dist (X s omega) (X t omega)) ^ a)
  dmu <=
  ENNReal.ofReal (C * ||s - t|| ^ ((d : R) + b))) :
  exists Y : (Fin d -> R) -> Omega -> S,
  (forall t, X t =^m[mu] Y t) /\
  forall p : R, forall hp : p in Ioo 0 (b / a),
  forall^m omega dmu, IsLocallyHolder <p, hp.1.le> (fun t => Y t
  omega) := by
sorry
```

Theorem 5.25 (portmanteau theorem, Alexandrov) Let $x_1, x_2, \dots$ be random elements in a metric space (S, ρ) with classes G, F of open and closed sets. Then these conditions are equivalent:

- (i) $x_n \rightarrow x$ in distribution,
- (ii) $\liminf_{n \rightarrow \infty} P\{x_n \in G\} \geq P\{x \in G\}$, G in $\mathcal{G}$,
- (iii) $\limsup_{n \rightarrow \infty} P\{x_n \in F\} \leq P\{x \in F\}$, F in $\mathcal{F}$,
- (iv) $P\{x_n \in B\} \rightarrow P\{x \in B\}$, B in $\mathcal{S}_x$.

For a random element x in a metric space S with Borel sigma-

Probability

```
theorem kallenberg_5_25_portmanteau
  {Omega Omega' S : Type*} [MeasurableSpace Omega] [MeasurableSpace Omega']
  [MetricSpace S] [MeasurableSpace S] [BorelSpace S]
  (mu : Measure Omega) (mu' : Measure Omega') [IsProbabilityMeasure mu]
  [IsProbabilityMeasure mu'] (xin : N -> Omega -> S) (xi : Omega' -> S)
  (hxi : forall n, AEMeasurable (xin n) mu) (hxi : AEMeasurable xi mu') :
  let lawsConverge := TendstoInDistribution xin atTop xi (fun _ => mu) mu'
  let openLowerBound := forall G : Set S, IsOpen G ->
  mu' (xi ^-1' G) <= liminf (fun n => mu (xin n ^-1' G)) atTop
```

Natural language statement

Domain

GroundTruth

field S , let S_{xi} denote the class of sets B in S with xi notin dB a.s., called the xi -continuity sets.

Theorem 5.27 (continuous mapping, Mann & Wald, Prohorov, Rubin) For any metric spaces S , T and set C subset S , consider some measurable functions $f, f_1, f_2, \dots : S \rightarrow T$ satisfying $sn \rightarrow s$ in $C \Rightarrow fn(sn) \rightarrow f(s)$. Then for any random elements $xi, xi_1, xi_2, \dots$ in S , $xin \rightarrow d xi$ in C a.s. $\Rightarrow fn(xin) \rightarrow d f(xi)$. In particular, we see that if $f : S \rightarrow T$ is a.s. continuous at xi , then $xin \rightarrow d xi \Rightarrow f(xin) \rightarrow d f(xi)$.

Probability

```
theorem kallenberg_5_27_continuous_mapping
{Omega Omega' S T : Type*} [MeasurableSpace Omega] [MeasurableSpace
Omega']
[MetricSpace S] [MeasurableSpace S] [BorelSpace S]
[MetricSpace T] [MeasurableSpace T] [BorelSpace T]
(mu : Measure Omega) (mu' : Measure Omega') [IsProbabilityMeasure mu]
[IsProbabilityMeasure mu'] (C : Set S) (f : S -> T) (fn : N -> S -> T)
(hfn : forall n, Measurable (fn n)) (hf : Measurable f)
(hcontinuous : forall s : S, s in C -> forall sn : N -> S,
  Tendsto sn atTop (nhds s) -> Tendsto (fun n => fn n (sn n)) atTop
    (nhds (f s)))
(xin : N -> Omega -> S) (xi : Omega' -> S)
(hxi : TendstoInDistribution xin atTop xi (fun _ => mu) mu')
(hC : mu' (xi ^-1' C) = 1) :
TendstoInDistribution (fun n omega => fn n (xin n omega)) atTop (fun
omega => f (xi omega))
  (fun _ => mu) mu' /\
  forall g : S -> T, Measurable g ->
    mu' (xi ^-1' {s : S | ContinuousAt g s}) = 1 ->
      TendstoInDistribution (fun n omega => g (xin n omega)) atTop (fun
        omega => g (xi omega))
    (fun _ => mu) mu' := by
sorry
```

Theorem 6.13 (Gaussian variance criteria, Lindeberg, Feller) Let (x_{ij}) be a triangular array with $E x_{ij} = 0$ and $\sum_j \text{Var}(x_{ij}) \rightarrow 1$, and let ζ be $N(0, 1)$. Then these conditions are equivalent:
(i) $\sum_j x_{ij} \rightarrow \zeta$ in distribution and $\sup_j \text{Var}(x_{ij}) \rightarrow 0$,
(ii) $\sum_j E(x_{ij}^2; |x_{ij}| > \epsilon) \rightarrow 0$, $\epsilon > 0$.
Here (ii) is the celebrated Lindeberg condition.

Probability

```
theorem kallenberg_6_13_gaussian_variance_criteria
{Omega Omega' : Type*} [MeasurableSpace Omega] [MeasurableSpace Omega']
(mu : Measure Omega) (mu' : Measure Omega') [IsProbabilityMeasure mu]
[IsProbabilityMeasure mu'] (k : N -> N)
(xi : (n : N) -> Fin (k n + 1) -> Omega -> R) (zeta : Omega' -> R)
(hximeas : forall n j, AEMeasurable (xi n j) mu)
(hxisq : forall n j, MemLp (xi n j) 2 mu)
(hcentered : forall n j, integral omega, xi n j omega dmu = 0)
(hindep : forall n, iIndepFun (xi n) mu)
(hvariance : Tendsto (fun n => sum j, variance (xi n j) mu) atTop (nhds
1))
(hzeta : HasLaw zeta (gaussianReal 0 1) mu') :
let sumConverges := TendstoInDistribution (fun n omega => sum j, xi n j
omega) atTop zeta
  (fun _ => mu) mu'
let maximalVarianceVanishes := Tendsto
  (fun n => Finset.univ.sup' (Finset.univ_nonempty_iff.mpr
    inferInstance)
    fun j => variance (xi n j) mu) atTop (nhds 0)
let lindeberg := forall eps : R, 0 < eps -> Tendsto
  (fun n => sum j, integral omega, (xi n j omega) ^ 2 * (Set.indicator
    {x : R | eps < |x|} (fun _ => (1 : R))) (xi n j omega) dmu) atTop
    (nhds 0)
  (sumConverges /\ maximalVarianceVanishes) <=> lindeberg := by
sorry
```

Theorem 8.5 (conditional distributions, disintegration) Let xi, η be random elements in S, T , where T is Borel. Then $L(xi, \eta) = L(xi)$ tensor μ for a probability kernel $\mu : S \rightarrow T$, where μ is unique a.e. $L(xi)$ and satisfies
(i) $L(\eta | xi) = \mu(xi, \cdot)$ a.s.,
(ii) $E(f(xi, \eta) | xi) = \int f(xi, t) \mu(dt)$ a.s., $f \geq 0$.

For any random element xi in a measurable space (S, S) , we define a conditional distribution of η , given xi , as a random measure of the form $\mu(xi, B) = P(\eta \in B | xi)$ a.s., B in T , for a probability kernel $\mu : S \rightarrow T$.

Probability

```
theorem kallenberg_8_5_conditional_distributions
{Omega S T : Type*} [MeasurableSpace Omega] [MeasurableSpace S]
[MeasurableSpace T] [StandardBorelSpace T] (mu : Measure Omega)
[IsProbabilityMeasure mu] (xi : Omega -> S) (eta : Omega -> T)
(hxi : Measurable xi) (heta : Measurable eta) :
exists kappa : Kernel S T, IsMarkovKernel kappa /\
  mu.map (fun omega => (xi omega, eta omega)) = mu.map xi tensor m
  kappa /\
  (forall kappa' : Kernel S T, IsMarkovKernel kappa' ->
    mu.map (fun omega => (xi omega, eta omega)) = mu.map xi tensor m
      kappa' -> kappa =^m[mu.map xi] kappa') /\
  forall f : S -> T -> R=>0inf, Measurable (Function.uncurry f) ->
```

Natural language statement

Domain

GroundTruth

		<pre> forall A : Set S, MeasurableSet A -> integral^~ omega in xi ^-1' A, f (xi omega) (eta omega) dmu = integral^~ omega in xi ^-1' A, integral^~ t, f (xi omega) t dkappa (xi omega) dmu := by sorry </pre>
Theorem 9.30 (optional sampling and closure, Doob) Let X be an F-submartingale on R^+, where X and F are right-continuous, and consider some optional times σ, τ, where τ is bounded. Then X_τ is integrable, and $X_{\sigma/\tau} \leq E(X_\tau \mathcal{F}_\sigma)$ a.s. This extends to unbounded times τ iff X^+ is uniformly integrable.	Martingales	<pre> theorem kallenber9_30_optional_sampling_and_closure {Omega : Type*} [MeasurableSpace Omega] (mu : Measure Omega) [IsProbabilityMeasure mu] (F : Filtration R>=0 MeasurableSpace Omega) [F.IsRightContinuous] (X : R>=0 -> Omega -> R) (hX : Submartingale X F mu) (hXright : forall^m omega dmu, forall t, ContinuousWithinAt (fun s => X s omega) (Ici t) t) (sigma tau : Omega -> WithTop R>=0) (hsigma : IsStoppingTime F sigma) (htau : IsStoppingTime F tau) (htaubounded : exists u : R>=0, forall^m omega dmu, tau omega <= u) : Integrable (stoppedValue X tau) mu /\ stoppedValue X (fun omega => min (sigma omega) (tau omega)) <=^m[mu] mu[stoppedValue X tau hsigma.measurableSpace] /\ (UniformIntegrable (fun t omega => max (X t omega) 0) 1 mu <-> forall tau' : Omega -> WithTop R>=0, IsStoppingTime F tau' -> Integrable (stoppedValue X tau') mu /\ forall sigma' : Omega -> WithTop R>=0, forall hsigma' : IsStoppingTime F sigma', stoppedValue X (fun omega => min (sigma' omega) (tau' omega)) <=^m[mu] mu[stoppedValue X tau' hsigma'.measurableSpace]) := by sorry </pre>
Theorem 10.5 (Doob-Meyer decomposition, Meyer, Doleans) For an adapted process X, these conditions are equivalent: (i) X is a local sub-martingale, (ii) $X = M + A$ a.s. for a local martingale M and a locally integrable, non-decreasing, predictable process A with $A_0 = 0$. The processes M and A are then a.s. unique. By an increasing process we mean a non-decreasing, right-continuous, and adapted process A with $A_0 = 0$. It is said to be integrable if $E A_\infty < \infty$. Recall that all sub-martingales are taken to be right-continuous. Local sub-martingales and locally integrable processes are defined by localization, in the usual way. A process M is said to be a local martingale, if it is adapted to F and such that the stopped and centered processes $M^{[taun]}$ - M_0 are true martingales for some optional times $taun$ coe inf. By a similar localization we may define local L^2-martingales, locally bounded martingales, locally integrable processes, etc. The required optional times $taun$ are said to form a localizing sequence.	Martingales	<pre> def LocalizesToInfinity {Omega iota : Type*} [MeasurableSpace Omega] [LinearOrder iota] [TopologicalSpace iota] [OrderTopology iota] (tau : N -> Omega -> WithTop iota) (mu : Measure Omega) : Prop := (forall n, tau n <= tau (n + 1)) /\ forall^m omega dmu, Tendsto (fun n => tau n omega) atTop atTop def IsLocalMartingale {Omega iota : Type*} [MeasurableSpace Omega] [LinearOrder iota] [Nonempty iota] [OrderBot iota] [TopologicalSpace iota] [OrderTopology iota] (X : iota -> Omega -> R) (F : Filtration iota MeasurableSpace Omega) (mu : Measure Omega) : Prop := exists tau : N -> Omega -> WithTop iota, (forall n, IsStoppingTime F (tau n)) /\ LocalizesToInfinity tau mu /\ forall n, Martingale (fun t omega => stoppedProcess X (tau n) t omega - X (bot : iota) omega) F mu def IsLocalSubmartingale {Omega iota : Type*} [MeasurableSpace Omega] [LinearOrder iota] [Nonempty iota] [OrderBot iota] [TopologicalSpace iota] [OrderTopology iota] (X : iota -> Omega -> R) (F : Filtration iota MeasurableSpace Omega) (mu : Measure Omega) : Prop := exists tau : N -> Omega -> WithTop iota, (forall n, IsStoppingTime F (tau n)) /\ LocalizesToInfinity tau mu /\ forall n, Submartingale (fun t omega => stoppedProcess X (tau n) t omega - X (bot : iota) omega) F mu def IsLocallyIntegrableProcess {Omega iota : Type*} [MeasurableSpace Omega] [LinearOrder iota] [Nonempty iota] [TopologicalSpace iota] [OrderTopology iota] (X : iota -> Omega -> R) (F : Filtration iota MeasurableSpace Omega) (mu : Measure Omega) : Prop := exists tau : N -> Omega -> WithTop iota, (forall n, IsStoppingTime F (tau n)) /\ LocalizesToInfinity tau mu /\ forall n t, Integrable (stoppedProcess X (tau n) t) mu theorem kallenber10_5_doob_meyer </pre>

Natural language statement

Domain

GroundTruth

		<pre> {Omega : Type*} [MeasurableSpace Omega] (mu : Measure Omega) [IsProbabilityMeasure mu] (F : Filtration R>=0 MeasurableSpace Omega) [F.IsRightContinuous] (X : R>=0 -> Omega -> R) (hX : Adapted F X) (hXright : forall^m omega dmu, forall t, ContinuousWithinAt (fun s => X s omega) (Ici t) t) : IsLocalSubmartingale X F mu <=> exists M A : R>=0 -> Omega -> R, (forall t, X t =^m[mu] M t + A t) /\ IsLocalMartingale M F mu /\ IsLocallyIntegrableProcess A F mu /\ IsStronglyPredictable F A /\ (forall^m omega dmu, Monotone fun t => A t omega) /\ (forall^m omega dmu, forall t, ContinuousWithinAt (fun s => A s omega) (Ici t) t) /\ A 0 =^m[mu] 0 /\ forall M' A' : R>=0 -> Omega -> R, (forall t, X t =^m[mu] M' t + A' t) -> IsLocalMartingale M' F mu -> IsLocallyIntegrableProcess A' F mu -> IsStronglyPredictable F A' -> (forall^m omega dmu, Monotone fun t => A' t omega) -> (forall^m omega dmu, forall t, ContinuousWithinAt (fun s => A' s omega) (Ici t) t) -> A' 0 =^m[mu] 0 -> (forall t, M t =^m[mu] M' t) /\ forall t, A t =^m[mu] A' t := by sorry </pre>
Theorem 23.2 (tightness and relative compactness, Prohorov) For a set X of random elements in a metric space S, we have (i) $\Rightarrow$ (ii) with (i) X is tight, (ii) X is relatively compact in distribution, and equivalence holds when S is separable and complete.	Probability	<pre> theorem kallenberg_23_2_tightness_and_relative_compactness {S : Type*} [MetricSpace S] [MeasurableSpace S] [BorelSpace S] (Xi : Set (ProbabilityMeasure S)) : let tight := IsTightMeasureSet {((nu : ProbabilityMeasure S) : Measure S) nu in Xi} let relativelyCompact := IsCompact (closure Xi) (tight -> relativelyCompact) /\ ((TopologicalSpace.SeparableSpace S /\ CompleteSpace S) -> (tight <-> relativelyCompact)) := by sorry </pre>
Theorem 23.6 (functional central limit theorem, Donsker) Let $x_1, x_2, \dots$ be i.i.d. random vectors in $\mathbb{R}^d$ with mean 0 and covariances $\text{Cov}(x_i, x_j) = \delta_{ij} \sigma^2$, form the continuous processes $X_t^n = n^{-1/2} \sum_{k=1}^{\lfloor nt \rfloor} x_k$, $t \geq 0$, $n \in \mathbb{N}$, and let B be a Brownian motion in $\mathbb{R}^d$. Then $X^n \rightarrow B$ in distribution in $C_0([0, \infty), \mathbb{R}^d)$. A d-dimensional Brownian motion is a process $B = (B_1, \dots, B_d)$ in $\mathbb{R}^d$, where $B_1, \dots, B_d$ are independent, one-dimensional Brownian motions.	Stochastic processes	<pre> def IsBrownianVector {Omega : Type*} [MeasurableSpace Omega] {d : N} (B : Omega -> C(R>=0, Fin d -> R)) (mu : Measure Omega) : Prop := (forall i, IsBrownianReal (fun t omega => B omega t i) mu) /\ forall times : Finset R>=0, iIndepFun (fun i omega (t : times) => B omega t i) mu theorem kallenberg_23_6_functional_central_limit {Omega Omega' : Type*} [MeasurableSpace Omega] [MeasurableSpace Omega'] {d : N} [MeasurableSpace C(R>=0, Fin d -> R)] [BorelSpace C(R>=0, Fin d -> R)] (mu : Measure Omega) (mu' : Measure Omega') [IsProbabilityMeasure mu] [IsProbabilityMeasure mu'] (xi : N -> Omega -> (Fin d -> R)) (X : N -> Omega -> C(R>=0, Fin d -> R)) (B : Omega' -> C(R>=0, Fin d -> R)) (hximeas : forall n, AEMeasurable (xi n) mu) (hiid : iIndepFun xi mu /\ forall n, IdentDistrib (xi n) (xi 0) mu mu) (hsecondMoment : forall i, MemLp (fun omega => xi 0 omega i) 2 mu) (hmean : forall i, integral omega, xi 0 omega i dmu = 0) (hcovariance : forall i j, integral omega, xi 0 omega i * xi 0 omega j dmu = if i = j then 1 else 0) (hx : forall n, 0 < n -> forall omega t i, X n omega t i = (Real.sqrt (n : R))^(i-1) * ((sum k in Finset.Icc 1 (n : R) * (t : R)+, xi k omega i) + ((n : R) * (t : R) - (n : R) * (t : R)+ * xi ((n : R) * (t : R)+ + 1) omega i))) (hxmeas : forall n, AEMeasurable (X n) mu) (hB : IsBrownianVector B mu') : TendstoInDistribution X atTop B (fun _ => mu) mu' := by sorry </pre>